

EVALUATING THE QUALITY OF LLM-GENERATED SPARQL QUERIES

FIFTH MEETING

Yassine Trabelsi

FIFTH TASK

What was done :

- Define evaluation metrics (SPINACH benchmark) using:

Ground truth queries :<https://github.com/ad-freiburg/grasp/blob/main/data/benchmark/wikidata/spinach/test.jsonl>

Predicted queries (LLM output) :<https://github.com/ad-freiburg/grasp/blob/main/data/benchmark/wikidata/spinach/outputs/gpt-5-mini.all.evaluation.json>

Steps:

- Extract properties from:
 - ground truth query
 - predicted query
- Compare the two sets of properties and define a metric:
- Property correctness score :
 - 1 $\rightarrow$ properties match
 - 0 $\rightarrow$ properties do not match

Evaluation Workflow: Property-Based Analysis (SPINACH Benchmark)

Input

- Ground truth queries → test.jsonl
- Predicted queries → LLM output (gpt-5-mini.all.evaluation.json)

Step 1 — Property extraction

- Extract properties using regex.
- Applied on:
 1. Ground truth queries
 2. Predicted queries

Step 2 — Merge datasets

Match queries using test_id

Output:

spinach_ground_truth_and_prediction_with_properties.csv

	A	B	C	D	E	F
	test_id	Question	SPARQL query	Extracted_properties_str	Predicted SPARQL query	Predicted_extracted_properties_str
1	test_0	Which distinct items are subclasses of execution methods or the subclasses of those, and so on?	PREFIX wdt: <http://www.wikidata.org/prop/direct/> PREFIX wd: <http://www.wikidata.org/entity/> SELECT DISTINCT ?class WHERE { ?class wdt:P279 + wd:Q15169167 . }	P279	PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX wd: <http://www.wikidata.org/entity/> PREFIX wdt: <http://www.wikidata.org/prop/direct/> SELECT DISTINCT ?class	P279

spinach_ground_truth_and_prediction_with_properties.csv

Step 3 — Property comparison

Compare sets of properties:

- Exact match → score = 1
- Otherwise → score = 0

test_id	Question	SPARQL query	Extracted_properties_str	Predicted SPARQL query	Predicted_extracted_properties_str	Property_correctness_score
1	test_0	PREFIX wdt: <http://www.wikidata.org/prop/direct/> PREFIX wd: <http://www.wikidata.org/entity/> SELECT DISTINCT ?class	P279	PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX wd: <http://www.wikidata.org/entity/> PREFIX wdt: <http://www.wikidata.org/prop/direct/> SELECT DISTINCT ?class	P279	1

spinach_property_evaluation.csv

Error Analysis

Step 4 — Retrieve all Wikidata properties + labels

→ avoid UNKNOWN values

	property	propertyLabel
1		
2	P10	video
3	P101	field of work
4	P102	member of political party
5	P103	native language
6	P105	taxon rank
7	P106	occupation
8	P108	employer
9	P109	signature
10	P110	illustrator

ALL_wikidata_properties_labels.csv

Step 5 — Property labeling

- Map each property to its label
- (e.g., P279 → subclass of)

Step 6 — Global performance

- Compute distribution of scores (0 vs 1)
- Visualization: correctness percentage

Step 7 — Error analysis (focus on score = 0)

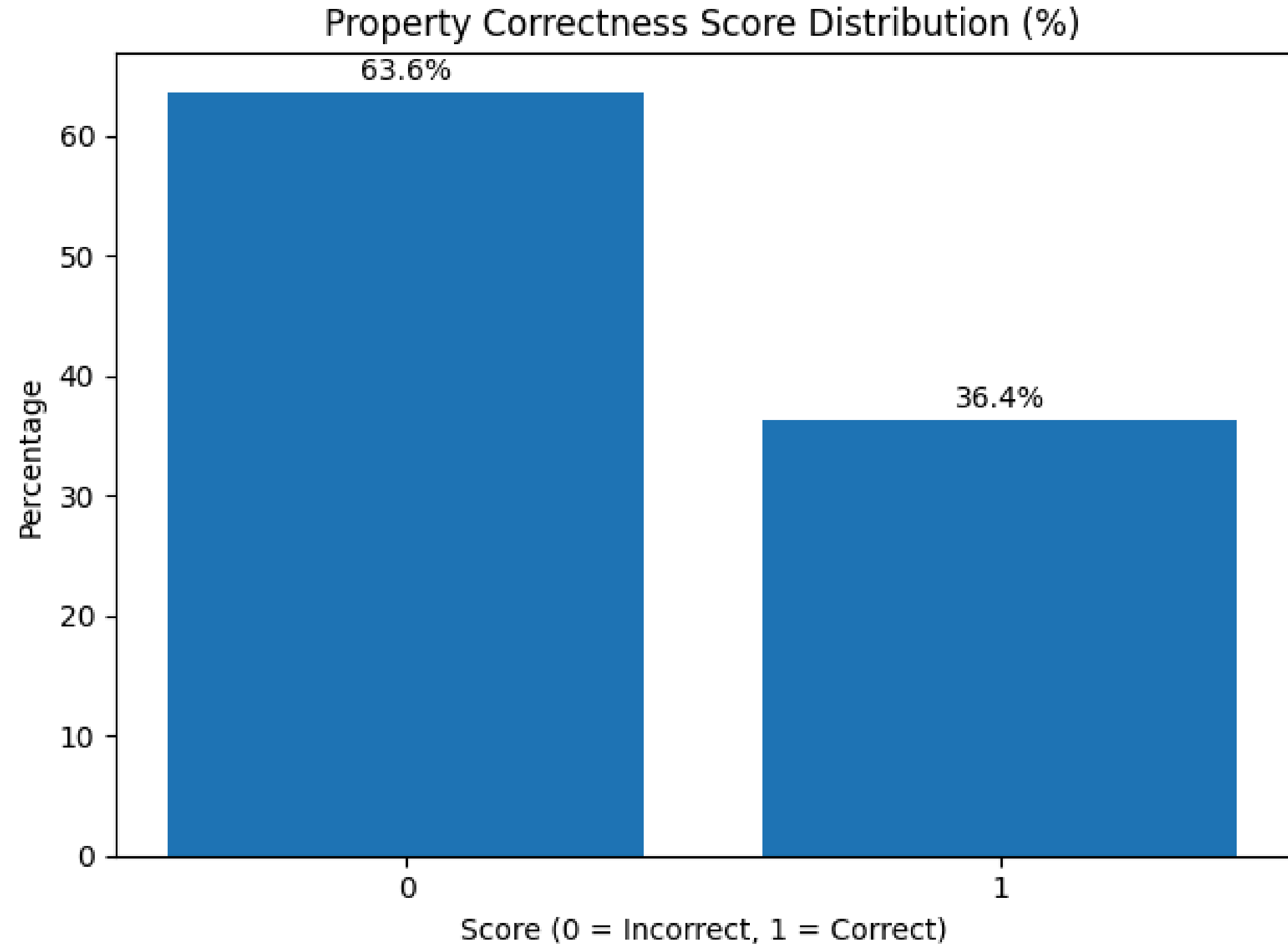
- Identify:
 - Missing properties (GT only)
 - Extra properties (Predicted only)
- Count frequency of each error type

Final outputs

test_id	Question	SPARQL query	Extracted_properties_str	Predicted SPARQL query	Predicted_extracted_properties_str	Property_correctness_score	All_properties_with_labels
test_0	Which distinct items are subclasses of execution methods or the subclasses of those, and so on?	PREFIX wdt: <http://www.wikidata.org/prop/direct/> PREFIX wd: <http://www.wikidata.org/entity/> SELECT DISTINCT ?class WHERE { ?class wdt:P279 + wd:Q15169167 . }	P279	PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX wd: <http://www.wikidata.org/entity/> PREFIX wdt: <http://www.wikidata.org/prop/direct/> SELECT DISTINCT ?item ?itemLabel WHERE { ?item wdt:P279 + wd:Q15169167 . OPTIONAL { ?item rdfs:label	P279	1	P279=subclasses of

spinach_property_evaluation_with_labels.csv

Final outputs



Final outputs

